

Ellipses and Hyperbolas: Reading a Model's Key Features

Vertices and foci of ellipses, vertices and foci of hyperbolas, and the asymptotes of a hyperbola. Every problem below gives you a real object modelled by a conic in standard position, with the units stated in the problem.

- For an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with $a > b$: vertices $(\pm a, 0)$, foci $(\pm c, 0)$, and $c^2 = a^2 - b^2$.
- For a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$: vertices $(\pm a, 0)$, foci $(\pm c, 0)$, $c^2 = a^2 + b^2$, and asymptotes $y = \pm \frac{b}{a}x$.
- Keep the units of the model in your answer, and show the arithmetic that produced each number.

1. The floor of a whispering gallery is an ellipse. With the centre of the room at the origin and both distances measured in feet, the wall follows $\frac{x^2}{169} + \frac{y^2}{144} = 1$. Two people standing at the foci can hear each other whisper. How far from the centre of the room is each focus?

Answer: _____ ft

2. Two radio towers stand on a hyperbola of positions given by $\frac{x^2}{36} - \frac{y^2}{64} = 1$, where x and y are in kilometres. Give the coordinates of the two vertices and of the two foci of this hyperbola.

Answer: _____

- 3.** A lawn sprinkler wets an elliptical patch described by $\frac{(x-3)^2}{25} + \frac{(y+2)^2}{9} = 1$, with x and y measured in metres from a corner stake. Give the coordinates of the two endpoints of the major axis, then the coordinates of the two foci.

Answer: _____

- 4.** The narrow waist of a cooling tower is modelled by the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = 1$, with x and y in metres. Far above and far below the waist the tower wall follows the asymptotes almost exactly. Find the slope of each asymptote, then write the equations of both asymptotes.

Answer: _____

5. A satellite travels an elliptical orbit with the centre of its planet at one focus. Its closest approach to that focus measures 4200 kilometres and its farthest distance measures 9800 kilometres. The closest distance equals $a - c$ and the farthest equals $a + c$, where a is the semi-major axis and c is the centre-to-focus distance. Write a system of two equations and solve it for a and c .

Answer: _____

6. Two navigation stations lie 20 kilometres apart, at the two foci of a hyperbola on which a ship sits. The ship's distances to the two stations differ by 12 kilometres, which is the vertex-to-vertex distance $2a$. Find a , then find b^2 for the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ that models the ship's possible positions.

Answer: _____

7. A hyperbolic mirror is ground so that its cross-section has vertices $(\pm 5, 0)$, with distances in centimetres, and asymptotes $y = \pm \frac{2}{5}x$. Find b , then find the centre-to-focus distance c , rounded to two decimal places.

Answer: _____

8. An architect models a bridge arch as the upper half of an ellipse centred at the origin, with x and y in metres. The survey shows the arch passes through the point $(3, 4)$ and through the point $(6, 2)$. Substituting each point into $\frac{x^2}{m} + \frac{y^2}{n} = 1$ gives a system in m and n ; solve it for $m = a^2$ and $n = b^2$, then give the coordinates of the two foci.

Answer: _____

- 9.** A tunnel cross-section follows $\frac{y^2}{49} - \frac{x^2}{16} = 1$, with x and y in metres. Jae reports that the asymptotes have slopes ± 0.57 . Explain what Jae did wrong, then give the correct slope of each asymptote.

Answer: _____

- 10.** A skate ramp's edge follows one branch of $\frac{x^2}{36} - \frac{y^2}{25} = 1$, with x and y in metres. **(a)** Find the slope of each asymptote. **(b)** On the grid below, draw the central rectangle, then both asymptotes, then sketch both branches. **(c)** Explain why the branches keep getting closer to the asymptotes without ever meeting them.

